Motivation Letter

Dear Members of the Selection Committee,

I still remember my summer vacation after completing my 10th standard, when I joined a local computer training class to learn the basics of computers. One day, we were introduced to **HTML**, and I created a simple webpage displaying my own name. Watching it appear in a web browser after writing just a few lines of code sparked a curiosity that has stayed with me ever since. It was a small achievement, but it made me wonder how a computer could transform a few lines of text into something meaningful.

That experience led me to choose the **Computer Science** group in higher secondary school. My interest soon expanded beyond programming to understanding how computers worked internally. I began exploring computer hardware, learning how RAM, storage, processors, and motherboards function together. Over time, I taught myself to assemble computers, troubleshoot hardware problems, and install operating systems from scratch. Looking back, understanding hardware made me appreciate that software and hardware are not separate worlds - they complement each other.

During my **Bachelor's in Information Technology** at Kongu Arts and Science College, that curiosity evolved into a passion for **software development**. Starting with HTML, CSS, and JavaScript, I developed websites and eventually built my own **portfolio website** from scratch. Seeing a complete website that I had designed and deployed myself was a defining moment. It showed me that programming was not simply about writing code but about transforming ideas into useful solutions. That realization motivated me to explore increasingly complex software systems.

My **Master's in Computer Applications** marked another important step in my journey. My perspective shifted from learning programming languages to understanding how software systems are designed and developed. Courses in **machine learning**, software engineering, database management, and web development introduced me to the relationship between data, algorithms, and software. One of my earliest projects involved **sentiment analysis**, where I developed a model capable of classifying social media posts as positive or negative based on textual content. Through this project, I gained an appreciation for how computational models can identify meaningful patterns from data.

Outside the classroom, I worked on software projects and explored embedded systems using Arduino and sensors, strengthening my understanding of software architecture, cloud deployment, and the interaction between software and hardware. By the end of my master's programme, I had grown into someone capable of designing complete technological solutions.

What if the software I built could solve problems beyond computers? What if it could directly improve people's lives?

That question led me to pursue a **Master in Medical Technology** at the **Indian Institute of Technology Jodhpur**, jointly offered with **AIIMS Jodhpur**. Entering an interdisciplinary environment where engineers, clinicians, and biomedical researchers worked together completely changed my perspective. I realized that developing healthcare technologies requires much more than technical knowledge. It requires understanding biological systems, clinical practice, experimental research, and the real challenges faced by patients and healthcare professionals.

My first step into this field was through **Surface-Enhanced Raman Spectroscopy (SERS)**. Initially, Raman spectra appeared to be nothing more than collections of peaks and noise. As I developed computational workflows for spectral preprocessing, baseline correction, feature extraction, and visualization, I gradually began to understand how those signals could reveal molecular signatures associated with disease.

As my research progressed, I came to appreciate that developing biomedical diagnostic technologies extends far beyond obtaining experimental results. As a **Junior Research Fellow** at IIT Jodhpur, I develop **Surface-Enhanced Raman Spectroscopy (SERS) biosensors** for detecting **sepsis-related biomarkers**. This work has taught me to approach scientific problems systematically, collaborate across disciplines, and continuously refine solutions through experimentation and validation. More importantly, it reinforced my aspiration to develop intelligent diagnostic systems that combine reliable biosensing with artificial intelligence to support timely clinical decision-making.

Alongside spectroscopy research, I also developed an **electromyography (EMG)**-based human-computer interaction system using Arduino, BioAmp EXG sensors, and Unity. Designing algorithms for noisy physiological signals reinforced the importance of careful preprocessing, calibration, validation, and experimental reproducibility. Although these projects addressed different applications, they shared the same objective - transforming complex biological signals into information that people can understand and use.

Every stage of my academic journey has prepared me for the same goal. Programming introduced me to problem-solving, machine learning revealed the power of data, and Medical Technology showed me how engineering can directly improve healthcare. Today, I see computer science, engineering, and medicine not as separate disciplines but as complementary fields working together to solve complex clinical challenges.

My long-term ambition is to become an independent researcher developing **AI-driven medical technologies** for **early disease diagnosis** and **clinically interpretable decision-support systems**. I aspire to integrate biomedical sensing, spectroscopy, and artificial intelligence to develop technologies that clinicians can trust and that ultimately improve patient care.

The **HEAL-4WARD Doctoral Network** particularly attracts me because it brings together artificial intelligence, biomarker research, clinical medicine, and biomedical engineering to address one of the most challenging problems in healthcare. I am especially interested in **DC11** because it represents the natural continuation of my current research on sepsis-related biomarker detection toward AI-driven clinical decision support. I am inspired by **Professor Evangelos J. Giamarellos-Bourboulis's** translational research, which successfully connects immunology, biomarker discovery, and clinical medicine to improve outcomes for patients with severe infections. What excites me most about the programme is its multidisciplinary training and international collaboration. The planned secondment at the **University of Vigo** would deepen my understanding of SERS-based infection biomarker analysis, while the research at **IRCCS-FSL** would provide valuable insight into biofilms and bacterial extracellular vesicles. Together with the clinical research environment at the **National and Kapodistrian University of Athens**, these opportunities would enable me to integrate biomedical sensing, artificial intelligence, and clinical medicine while developing into an independent researcher capable of translating scientific discoveries into practical healthcare solutions.

From creating my first webpage as a school student to developing biosensors for sepsis-related biomarker detection, my journey has been driven by the same curiosity: understanding how technology can solve meaningful problems. The **DC11** project represents the natural continuation of that journey. It offers the opportunity to combine my background in computer science, biomedical engineering, and translational healthcare research while developing the expertise needed to build clinically interpretable AI systems for early diagnosis and personalised treatment of infectious diseases. I would be honoured to contribute to the HEAL-4WARD Doctoral Network while growing as an independent researcher committed to developing AI-enabled healthcare technologies that translate scientific discoveries into meaningful clinical impact.

Yours sincerely,

Sukesh Periyasamy